MILITARY SPECIFICATION

CASING, BURSTER M158

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements, quality assurance provisions, and packaging for the M158 Burster Casing for the 81MM Mortar WP round.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

10521905 - Casing, Burster, M158

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

PUBLICATIONS

American National Standards Institute (ANSI)

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018)

American Society of Testing Materials (ASTM)

ASTM Method E-8 - Tension Testing of Metallic Materials

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.1.3 Order of precedence . In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Material. Material shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on drawing (dwg.) 10521905 as applicable, and with all requirements specified in applicable specifications and standards.

3.3 Mechanical properties. Mechanical properties specified on the applicable drawing shall be uniform throughout each burster casing.

3.3.1 Metal defects. The bursting casing shall be free from cracks, splits, folds, cold shuts, inclusions, porosity or similar defects.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, sharp edges, chips, dirt, grease, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspections

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items as required per Table 1.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-50779A(AR)

PARAGRAPH	TITLE: Casing, Burster, M158		SHEET 1 OF 1		DRAWING NUMBER: See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Casing, Burster M158 (dwg. 10521905)				
	Examination for defects	10		3.2	4.4.2.1
	Tension Test	2		3.2	4.5.3
NOTES:					

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50779A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Casing, Burster, M158				10521905
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Diameter of press fit surface		100%	3.2	Gage
2	Leak test		100%	3.2	Test/4.5.2
3	Metal defective		100%	3.3.1	Visual
4	Finish of press fit surface improper		100%	3.2	Visual, see 6.4
Major					
101	Tension test	2 1/		3.2	Test/4.5.3
102	Runout of pitch diameter of thread with press fit surface, including front face		0.40%	3.2	Gage
103	Runout of rear of tube with press fit surface, including rear front of flange		0.40%	3.2	Gage
104	Perpendicularity of press fit surface with rear face of flange		0.40%	3.2	Gage
105	Straightness of tube inside		0.40%	3.2	Gage
106	Pitch diameter of thread, maximum		0.40%	3.2	Gage
107	Minor diameter of thread, maximum		0.40%	3.2	Gage
108	Length of effective thread, minimum		0.40%	3.2	Gage
109	Length of press fit surface		0.40%	3.2	Gage
110	Thickness of wall		0.40%	3.2	Gage
111	Taper at rear of press fit surface incorrect		0.40%	3.2	Gage
112	Profile of radius with press fit surface		0.40%	3.2	Gage

NOTES: 1/ Samples shall be selected from each 8 hours production or fraction thereof. Failure of either sample to comply shall be cause for rejection of the produciton represented by the sample.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50779A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Casing, Burster, M158				10521905
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Pitch diameter of thread, minimum		0.65%	3.2	Gage
202	Minor diameter of thread, minimum		0.65%	3.2	Gage
203	Diameter of flange		0.65%	3.2	Gage
204	Diameter of front face		0.65%	3.2	Gage
205	Diameter of tube inside, maximum		0.65%	3.2	Gage
206	Diameter of undercut, min. (if applicable)		0.65%	3.2	Gage
207	Depth of fuze cavity		0.65%	3.2	Gage
208	Depth of grooves		0.65%	3.2	Gage
209	Length from front face to rear of flange		0.65%	3.2	Gage
210	Length of fuze housing		0.65%	3.2	Gage
211	Length of thread, maximum		0.65%	3.2	Gage
212	Length, total		0.65%	3.2	Gage
213	Thickness through base		0.65%	3.2	Gage
214	Width of undercut (if applicable)		0.65%	3.2	Gage
215	Location of profile of grooves		0.65%	3.2	Gage
216	Radii or chamfers missing or incorrect		0.65%	3.2	Visual (see 6.4)
217	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the “paragraph Reference/Inspection Method” column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table 1.

4.5.2 Leak test. Each burster case shall be mounted in an approved fixture in such a manner that the burster case will be completely submerged in water, and then subjected to the required air pressure for the specified time duration. A burster casing shall be classed defective if evidence of bubbles rising to the surface of the water or appearing on any surface of the casing is observed.

4.5.3 Tension test. From the samples selected in accordance with 4.4.2.1, one full sectioned tension test specimen as prescribed by ASTM Method E-8 shall be taken from the tube position of each of the 2 burster casing samples. The test shall be peformed using equipment and procedures prescribed by ASTM Method E-8.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the requirements of Level B in MIL-STD-1169.

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6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 81MM Cartridge, Smoke, WP, M374 Series.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

6.4 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.5 Delay in distribution. Document was approved 9 November 1982, however, it was not received by Department of Defense Single Stock Point for indexing and distribution until December 1983.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-50779A	CASING, BURSTER M158

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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